GreenRoute

The team:

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Introduction:

The United Nations Sustainable Development Goal 13 calls for urgent action to combat climate change and its impacts by 2030 *(UNDP, Climate goals 2030)*. Climate change has become one of the most pressing challenges of our time. While it is often perceived as a distant issue that will mainly affect future generations, its consequences are already visible today. From slightly warmer winters to an increasing frequency of natural disasters, the effects are becoming more apparent each year. One of the main drivers of climate change is **carbon emissions**, with the transportation sector representing a significant share. Transport stands for 30% of the climate emissions in Norway *(Statistic Norway, 2019).* Everyday activities such as commuting to work, taking weekend trips, or travelling by plane contribute to these emissions.

To address the challenge of transport carbon emissions, we developed *GreenRoute*. This is an application designed to increase awareness of how everyday travel choices affect the environment. Many people are unaware of how much carbon different modes of transport produce, and this lack of information can make it difficult to make more sustainable decisions. *GreenRoute* aims to close this gap by providing users with clear and accessible information about the emissions associated with different travel options. By making this data visible and easy to interpret, the application helps users better understand the environmental consequences of their mobility choices.

The concept behind the name *GreenRoute* reflects the purpose of the application. We want to guide users toward more sustainable routes and transportation habits. Rather than only presenting raw data, the app focuses on communicating environmental impact in a simple and intuitive way so that users can easily compare alternatives and make informed decisions.

Skal vi her legge inn at vi ønsker å skille oss ut ved å komme med pushvarsler som gir ultimatum og spiller på følelser. "Er du klar over at hvis du bruker denne løsningen så dør 10 elefanter?" "100 barn mister rent drikkevann av denne løsningen"

By combining digital technology with clear visualization, *GreenRoute* seeks to encourage more sustainable behavior in everyday life. In this paper, we present the idea behind the application, explain how it is developed, and discuss how tools like *GreenRoute* can contribute to greater awareness of the climate impact of transportation.

Meet the team:

The development of GreenRoute has been carried out by a team of six students from the Norwegian School of Economics. The group includes students from Norway, Switzerland, France, and Spain, bringing together perspectives shaped by different cultural backgrounds. This international composition has helped us approach the project with a broader understanding of the challenges related to sustainable mobility.

Collaboration within the team has been an important part of the project process. At an early stage, we identified our individual strengths and distributed the tasks accordingly. This helped structure the work and ensured that different parts of the project progressed efficiently. Through discussion and cooperation, we refined our ideas and shaped the concept behind *GreenRoute*. This exchange of perspectives strengthened the overall outcome of the project.

During the development of the project, the team also faced an unexpected challenge. At one point in the process, roughly one third of the original group decided to withdraw from the project. This required the remaining members to reorganize the structure of the work and redistribute responsibilities. Although this situation initially created some uncertainty, it eventually strengthened the collaboration within the team. With fewer members, we experienced a stronger sense of commitment and responsibility, as those who remained were motivated to contribute actively and take part in the learning process throughout the project.

Methodology:

- General about methodology and what is our focus area

The application estimates carbon emissions by combining several variables related to transportation. These include factors such as travel distance, vehicle type, fuel type, and the number of passengers. By entering information about a specific trip, users are able to calculate the estimated emissions associated with their journey. The tool also allows users to compare different transportation options, making it easier to understand how travel choices influence environmental impact.

According to *The Intergovernmental Panel On Climate Change* carbon dioxide emission is calculated by the following formula:

$$CO_2 = d * (\frac{ef}{p})$$

d = distance travelled in km

ef = emission factor (based on the transport used)

p = number of passengers

For certain modes of transportation, including airplanes, buses, and ferries, emissions are estimated using the following formula:

$$CO_2 = d * ef$$

In this approach, passenger numbers are not included as a separate variable. This is because the emission factors already take into account average occupancy levels and typical operating conditions. According to the UK Department for Energy Security and Net Zero (DESNZ), these factors are designed to represent emissions per passenger for different transport modes.

Still using UK Department for Energy Security and Net Zero (DESNZ) as source:

Cars:

Electric cars depend on the source of electricity generation.

Gasoline car	Diesel car	Hybrid car	Electric car
0,171 kg CO_2e/km	0,171 kg CO_2e/km	0,110 kg CO_2e/km	0,050 kg CO_2e/km

- Formula (fremtiden I våre hender) https://www.framtiden.no/tips/klimagassutslippet-fra-ulike-reisemaater
    - Cars

- Bybanen (skyss)
- Bus
- Motorbike
- Train (vy.no)
- Ferry (ferry gogo)
- Airplane
- Walking and biking (obviously the best option)

Business model:

- General
- Costumer segment (who we want to reach out to,
- Costumer relationship and value
- How the app works
- Revenue

Cost for environment vs real monnez cost for the consumers (and also time it takes)

Create a problem (make aware of the pollution) but give solutions to what is what

We can aim at multiple people (general population AND BUSINESSES who wants for their employees to have a greener impact if they do business travel)

The "Emotional Push Notification" Angle

You mentioned this in the doc — this is actually your most original idea. Frame it as a **behavioral nudge system** inspired by behavioral economics (Thaler & Sunstein's nudge theory). Instead of just data, you make it *visceral*: "This trip produces enough CO_2 to melt X liters of Arctic ice." That's memorable and differentiates you from every other carbon calculator.

What makes us DIFFERENT --> Most carbon calculators give you a number. GreenRoute gives you a feeling and an alternative.

Clearly state

Business Model — Two Clear Segments

Present a **B2C + B2B dual model**: (2 different revenue models)

- B2C: Free app with premium features (detailed stats, monthly reports, gamification/badges)
  (more to gain attraction)

Environmentally Conscious Users

Individuals who want to track and reduce their personal carbon footprint by understanding the emissions generated by their travel choices.

Students and Young Professionals

Young, digitally engaged users who want a simple and interactive way to visualize the environmental impact of their everyday mobility.

Frequent Travelers

People who travel often and want to monitor the cumulative emissions of their trips and identify lower-emission alternatives.

- B2B: Sell to companies who need to track employee business travel emissions for **ESG reporting** — this is a *real and growing market* since EU's CSRD regulation now requires large companies to report Scope 3 emissions (which includes employee travel)
  (biggest revenue stream)

Large Companies with ESG Reporting Requirements

Organizations that must measure and report travel emissions due to regulations such as the **Corporate Sustainability Reporting Directive from the European Union.

Sustainability Consulting Firms

Consulting firms that advise companies on ESG strategies and can use the platform to measure and manage travel-related emissions for their clients.

Corporate Travel Management

Companies and travel management services that want to integrate carbon data into travel planning to show the true environmental cost of different transport options.

--> maybe collab (B2B case) where we show what is the true cost of the travel

Customer relationship and value

Customer relationships are a key feature to make this app work since it is not only important to attract them but also creating a bond for them not to leave the app. Since the app is created to influence everyday travel decisions, the most important relationships will be with users that make more than one interaction.

For **individuals**, GreenRoute is mainly going to rely on self-service relationships. Users can independently access information about the carbon emissions of different transport options, compare alternatives, and receive recommendations directly through the app. This creates an easy to use service with scalability since many users can find benefits by using it.

A key element of customer relationships in GreenRoute is user engagement through **emotional factors**. Instead of only showing numbers and scientific data, the app is going to show personalized messages depending on the choice that each user is looking forward to making. Moreover, the application can maintain user interest by sending notifications that make the environmental impact of travel choices feel more immediate and relevant.

In the **business-to-business** segment, customer relationships would be more structured and long-term. Companies using GreenRoute would expect a reliable and professional service that supports their sustainability objectives. In this case, the relationship would be based on the solutions the app provides for employee travel monitoring. This would create a more digital relationship where interactions are mainly done through the app.

Sources:

- FN: https://www.undp.org/sustainable-development-goals/climate-action
- SSB: https://www.ssb.no/natur-og-miljo/artikler-og-publikasjoner/transport-star-for-30-prosent-av-klimautslippene-i-norge#:~:text=Transport%20st%C3%A5r%20for%20nesten%20en%20tredjedel%20av,i%20Norge%2C%20if%C3%B8lge%20statistikken%20Utslipp%20til%20luft.
- IPCC: https://www.ipcc.ch/
- DESNZ :https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2023
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Environmentally Conscious Users

This group includes individuals who want to better understand and reduce their personal carbon footprint. The platform allows them to track the emissions generated by their travel and compare different transportation options.

A **free version** provides basic CO_2 calculations and travel tracking, while a **premium version** offers advanced features such as detailed statistics, monthly reports, and gamification elements like sustainability badges.

Students and Young Professionals

Students and young professionals are another key audience. They frequently use multiple transport options and are often interested in sustainable lifestyles.

For them, the platform provides a simple way to visualize the environmental impact of everyday mobility decisions. Gamification features and progress tracking increase engagement and encourage continued use.

Frequent Travelers

Frequent travelers and commuters can use the platform to track the cumulative emissions of long-distance travel, such as flights or long car trips. Premium features provide deeper insights into travel patterns and potential lower-emission alternatives.

B2B – Corporate Clients

Large Companies with ESG Reporting Requirements

Many large companies must now report environmental data due to regulations such as the **Corporate Sustainability Reporting Directive introduced by the **European Union**.

The platform enables organizations to track emissions from employee business travel. Companies receive aggregated data and reports that help them measure **Scope 3 emissions** and simplify ESG reporting.

Sustainability Consulting Firms

Sustainability and ESG consulting firms can integrate the platform into their services when helping clients measure and reduce travel-related emissions. This creates partnership opportunities and expands the platform's reach to multiple corporate clients.

Corporate Travel Management

Another opportunity is collaboration with corporate travel management systems. The platform can help companies see the **true environmental cost of travel options**, enabling more informed and sustainable travel decisions.